For k at most n, truncating a type to n and then to k is the same as truncating it to k

The Univalent Foundations Program

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Editorial note. The counted statement is the author lem environment at source lines 743-746 and its proof environment opens on line 747, the next line; the pairing is by adjacency and the retained proof range 747-764 is the whole environment.

Editorial note. The book label-number dump gives this label as 7.3.15 (env lem, chapter 7, section 3, fifteenth counter value), so the counters chapter 7, section 3 and thm 14 (zero-based) are set so that the item prints with the book number.

Editorial note. The counted statement and proof refer to no label outside the card: the retained ranges carry no cross reference command, and the truncation recursion principle is a standard prerequisite, not a delegation of a step of this proof.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: the collapse of an iterated truncation (author label lem:truncation-le, book Lemma 7.3.15).

Allowed standard prerequisites (editorial).

- Truncation and its recursion principle. The truncation of a type to a level is k-truncated itself and maps into it are defined by the recursion principle, which requires the target to be k-truncated.. Both maps of the equivalence are defined by that principle and their well-definedness rests on the truncation levels of their targets.
- n-type-ness is downward closed. A k-truncated type is also n-truncated whenever k is at most n.. It is the reason the first map is well defined.

Lemma 7.3.15. *Let $k, n \geq -2$ with $k \leq n$ and $A : \mathcal{U}$. Then $\|\|A\|_n\|_k = \|A\|_k$.*

Proof. We define two maps $f : \|\|A\|_n\|_k \to \|A\|_k$ and $g : \|A\|_k \to \|\|A\|_n\|_k$ by

$$f(||a|_n|_k) :\equiv |a|_k \qquad \text{and} \qquad g(|a|_k) :\equiv ||a|_n|_k.$$

The map f is well-defined because $\|A\|_k$ is k-truncated and also n-truncated (because $k \leq n$), and the map g is well-defined because $\|\|A\|_n\|_k$ is k-truncated.

The composition $f \circ g : \|A\|_k \to \|A\|_k$ satisfies $(f \circ g)(|a|_k) = |a|_k$, hence $f \circ g = \mathsf{id}_{\|A\|_k}$. Similarly, we have $(g \circ f)(||a|_n|_k) = ||a|_n|_k$ and hence $g \circ f = \mathsf{id}_{\|\|A\|_n\|_k}$. □